

Lab 7: Noise Reduction

Matlab comes with a speech data set of someone saying ‘Matlab’. The Matlab command `load mtlb` will load this data set. If your computer supports sound, you can listen to the original signal with the command `soundsc(mtlb)`. [Type `help soundsc` to find out how this command works.] The speech data set was obtained with a sampling frequency of $F_s = 7418$ Hz. The sampling period is $T_s = 1/F_s$.

The signal can be plotted with the following commands.

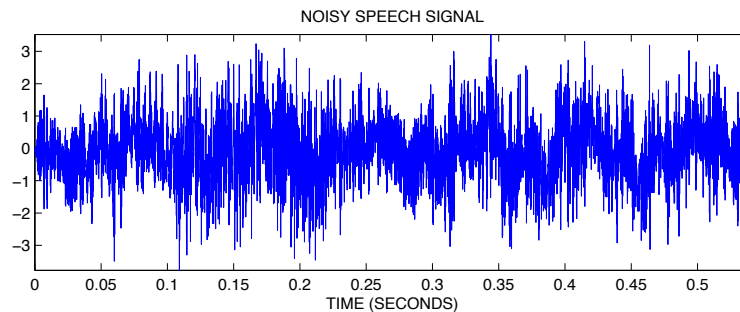
```
>> clear, close all
>> load mtlb
>> who
```

Your variables are:

```
Fs      mtlb
```

```
>> L = length(mtlb);
>> plot([1:L]/Fs,mtlb)
>> axis tight
>> xlabel('TIME (SECONDS)')
```

The data set `NoisySpeech.txt` on the class webpage, shown in the following figure, contains the `mtlb` data set corrupted by additive noise.



We will call the noisy signal $x(n)$. You are to design a low-pass filter to filter out the noise. You will try different low-pass filters until you find one that works well.

1. Download the data from the course webpage and make a plot of the data. You should get the same figure as shown above.

To obtain the data file, go to the course webpage and save the file `NoisySpeech.txt`, then type `load NoisySpeech.txt` on the Matlab command line. (You may need to change your Matlab working directory with the `cd` command to where-ever you have saved the file. Alternatively, you can append the directory to the Matlab path with the command `path`).

Typing `load NoisySpeech.txt` on the Matlab command line will create a vector called `NoisySpeech`. For convenience, type `x = NoisySpeech;` on the command line so that you can use `x` instead of the longer variable name. The command `plot([1:L]/Fs,x)` will then display the plot above.

If your computer supports sound, you can listen to the noisy signal $x(n)$ with the command `soundsc(x)`.

2. Use the `dtft` command as shown to plot the magnitude of the spectrum of the signal `mtlb`. What frequencies are predominant in the speech signal?

The `dtft` is not a built in Matlab command, you must download it from the course webpage. DTFT stands for the Discrete-Time Fourier Transform.

```
>> [M,f] = dtft(mtlb,1/Fs);  
>> plot(f,M)  
>> xlabel('FREQUENCY (Hz)')  
>> title('SPECTRUM of MTLB')
```

3. As in the previous part, use the `dtft` command to plot $|X^f(\omega)|$, the spectrum of the noisy signal $x(n)$.

Can you identify which part of the spectrum corresponds to the noise and which part corresponds to the speech signal?

Explain how a low-pass filter can remove the high frequency noise.

From the spectrum can you estimate what the cut-off frequency of the lowpass filter should be?

4. Design an appropriate elliptical low-pass filter with the `ellip` command. Use the `filter` command to filter the noisy signal $x(n)$ with the lowpass filter you design.

Try different cut-off frequencies and different values for the stop-band ripple and the pass-band ripple. Can you remove most of the noise from the noisy signal?

Is the noise reduction quality affected more by the size of passband ripple or by the size of the stopband ripple? (What is more important — that the passband be very close to 1 or that the stopband be very close to 0? You should try different filter designs and listen to the results to find out.)